

Lecture 3

3- Data Description

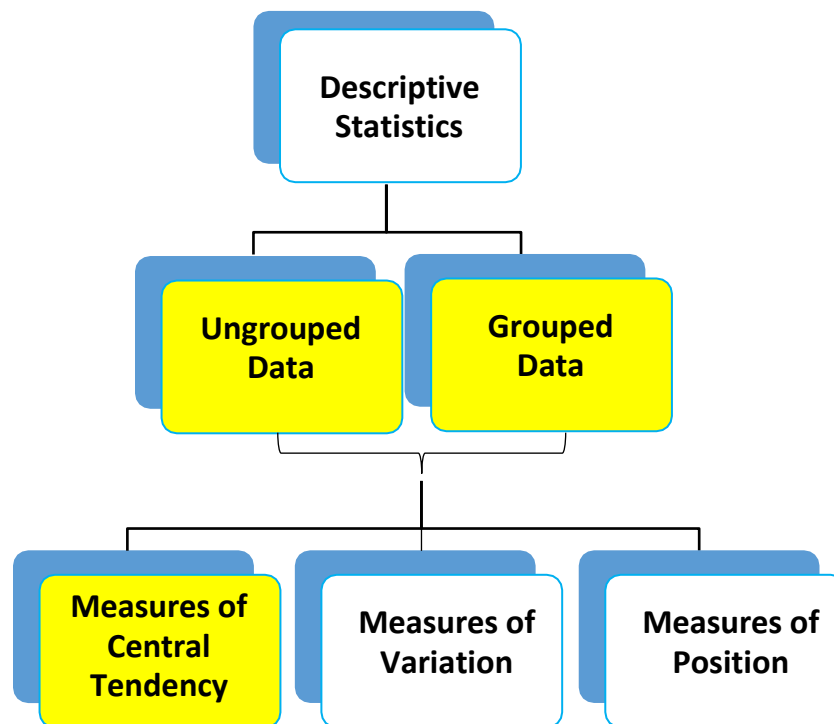
After completing the previous stages which are *collecting*, *organizing* and *presenting* data for statistics, **summarizing data is the next stage that can be done by various statistical methods.**

Statisticians use samples taken from populations; however, when populations are small, it is not necessary to use samples since the entire population can be used to gain information. Measures found by using all the data values in the population are called **parameters**. Measures obtained by using the data values from samples are called **statistics**.

A **statistic** is a characteristic or measure obtained by using the data values from a sample.

A **parameter** is a characteristic or measure obtained by using all the data values from a specific population.

(Roman letters will be generally used to indicate sample statistics and Greek letters to indicate population parameters).



3–1 Measures of Central Tendency

Arithmetic Mean

Case 1: Ungrouped Data

The *arithmetic mean*, also known as the *arithmetic average*, is found by adding the values of the data and dividing by the total number of values.

a) The **mean** for sample with size n , is given as:

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots}{n} \quad \frac{\sum_{i=1}^n X_i}{n}$$

where n represents the total number of values in the sample.

b) For a population with size N , the mean is given as:

$$\mu = \frac{X_1 + X_2 + X_3 + \dots}{N} = \frac{\sum_{i=1}^N X_i}{N}$$

where N represents the total number of values in the population.

Example 3–1: The data represent the number of days off per year for a sample of individuals selected from nine different countries. Find the mean.

20, 26, 40, 36, 23, 42, 35, 24, 30

Solution:

$$\bar{X} = \frac{\sum X}{n} = \frac{20 + 26 + 40 + 36 + 23 + 42 + 35 + 24 + 30}{9} = \frac{276}{9} = 30.7 \text{ days}$$

Hence, the mean of the number of days off is 30.7 days.

Case 2: Grouped Data

$$\bar{X} = \frac{\sum_{i=1}^n f_i \cdot X_i}{\sum_{i=1}^n f_i} = \frac{1}{n} \sum_{i=1}^n f_i \cdot X_i$$

X_i : is the midpoint of the class i ,

where f_i : is the frequency for the class i in the sample,

$n = \sum_{i=1}^n f_i$: is the sample size (sum of all frequencies).

Example 3–2: The following table gives the frequency distribution of the number of orders received each day during the past 50 days at the office of a mail-order company. Calculate the mean.

Solution

Number of orders	f
10 – 12	4
13 – 15	12
16 – 18	20
19 – 21	14

$n = 50$

Number of orders	f	X	$f \cdot X$
10 – 12	4	11	44
13 – 15	12	14	168
16 – 18	20	17	340
19 – 21	14	20	280
	$n = 50$		$= 832$

X is the midpoint of the class. It is obtained by adding the class limits and divide by 2.

$$\bar{X} = \frac{\sum f \cdot X}{n} = \frac{832}{50} = 16.64$$

The Median

Case 1: Ungrouped Data

The **median** is the middle value in an ordered sequence of data. The symbol for the median is MD.

The positioning-point formula $\frac{(n+1)}{2}$ used to find the place in the ordered array corresponds to the median value.

1. If there are an odd number of observations in the data set, the median is represented by the numerical value corresponding to the positioning point, the $\frac{(n+1)}{2}$ ordered observation.
2. If there are an even number of observations in the data set, then the positioning point lies between the two observations in the middle of the data set. The median is then the average of the numerical values corresponding to these two middle observations.

Example 3–3: The number of rooms in the seven hotels in downtown Pittsburgh is 713, 300, 618, 595, 311, 401, and 292. Find the median.

Solution:

Step 1 Arrange the data in order.

292, 300, 311, 401, 595, 618, 713

Step 2 Select the middle value.

292, 300, 311, 401, 595, 618, 713

↑
Median

Hence, the median is 401 rooms.

Example 3–4: The number of cloudy days for the top 10 cloudiest cities is shown. Find the median.
209, 223, 211, 227, 213, 240, 240, 211, 229, 212

Solution:

Arrange the data in order.

209, 211, 211, 212, 213, 223, 227, 229, 240, 240

↑
Median

$$MD = \frac{213 + 223}{2} = 218$$

Hence, the median is 218 days.

Case 2: Grouped Data

Step 1: Construct the cumulative frequency distribution.

Step 2: Decide the class that contain the median.

Class Median is the first class with the value of cumulative frequency equal at least $n/2$.

Step 3: Find the median by using the following formula:

$$\text{Median} = MD = L + i \left(\frac{\frac{n}{2} - F_{m-1}}{f_m} \right)$$

where:

n = the total frequency

F_{m-1} = the cumulative frequency of the class preceding the median class

f_m = the frequency of the class median

i = the class width in which median lies

L = the lower class boundary of the class in which median lies

Example 3–5: Based on the grouped data below, find the median.

Solution

Seconds	f
51 – 55	2
56 – 60	7
61 – 65	8
66 – 70	4

Construct the cumulative frequency distribution

Height (in cm)	f	cf
51 – 55	2	2
56 – 60	7	9
61 – 65	8	17
66 – 70	4	21

$$\frac{n}{2} = \frac{21}{2} = 10.5 \rightarrow \text{class median is the 3}^{rd} \text{ class}$$

So, $F_{m-1} = 9$, $f_m = 8$, $i = 5$ and $L = 60.5$

Therefore,

$$\begin{aligned} MD &= L + i \left(\frac{\frac{n}{2} - F_{m-1}}{f_m} \right) \\ &= 60.5 + 5 \left(\frac{10.5 - 9}{8} \right) \\ &= 60.5 + 0.9375 = 61.4375 \end{aligned}$$

The Mode

Case 1: Ungrouped Data

The value that occurs most often in a data set is called the **mode**.

- A data set that has only one value that occurs with the greatest frequency is said to be **unimodal**.
- If a data set has two values that occur with the same greatest frequency, both values are considered to be the mode and the data set is said to be **bimodal**.
- If a data set has more than two values that occur with the same greatest frequency, each value is used as the mode, and the data set is said to be **multimodal**.
- When no data value occurs more than once, the data set is said to have **no mode**.

Example 3–6: The data show the number of licensed nuclear reactors in the United States for a recent 15-year period. Find the mode.

104 104 104 104 104
 107 109 109 109 110
 109 111 112 111 109

Solution

Since the values 104 and 109 both occur 5 times, the modes are 104 and 109. The data set is said to be bimodal.

Case 2: Grouped Data

- The mode for grouped data is the modal class. The **modal class** is the class with the largest frequency.

$$Mode = L + i \left(\frac{f_m - f_{m-1}}{(f_m - f_{m-1}) + (f_m - f_{m+1})} \right)$$

where:

f_{m-1} = the frequency of the group before the modal group

f_{m+1} = the frequency of the group after the modal group

f_m = the frequency of the modal group

i = the group width

L = the lower class boundary of the modal group

Example 3–7: Based on the grouped data below, find the mode.

Solution

Seconds	f
51 – 55	2
56 – 60	7
61 – 65	8
66 – 70	4

$$L = 60.5, f_{m-1} = 7, f_m = 8, f_{m+1} = 4, i = 5$$

$$Mode = 60.5 + 5 \left(\frac{8 - 7}{(8 - 7) + (8 - 4)} \right)$$

$$= 60.5 + 5 \left(\frac{1}{5} \right) = 61.5 \text{ s}$$

The Midrange

The **midrange** is defined as the sum of the lowest and highest values in the data set, divided by 2. The symbol MR is used for the midrange.

$$MR = \frac{\text{lowest value} + \text{highest value}}{2}$$

Example 3–8: Find the midrange of data:

18.0, 14.0, 34.5, 10, 11.3, 10, 12.4, 10

Solution

The smallest bonus is \$10 million and the largest bonus is \$34.5 million.

$$MR = \frac{10 + 34.5}{2} = \frac{44.5}{2} = \$22.25 \text{ million}$$

Notice that this amount is larger than seven of the eight amounts and is not typical of the average of the bonuses. The reason is that there is one very high bonus, namely, \$34.5 million.

The Weighted Mean

Find the **weighted mean** of a variable X by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\bar{X} = \frac{w_1X_1 + w_2X_2 + \dots + w_nX_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum_{i=1}^n w_iX_i}{\sum_{i=1}^n w_i}$$

where $w_1, w_2, \dots, w_n$ are the weights and $X_1, X_2, \dots, X_n$ are the values.

Example 3–9: A student received an A in English Composition I (3 credits), a C in Introduction to Psychology (3credits), a B in Biology I (4credits), and a D in Physical Education (2credits). Assuming A =4 grade points, B =3 grade points, C =2 grade points, D =1 grade point, and F =0 grade points, find the student's grade point average.

Solution

Course	Credits (w)	Grade (X)
English Composition I	3	A (4 points)
Introduction to Psychology	3	C (2 points)
Biology I	4	B (3 points)
Physical Education	2	D (1 point)

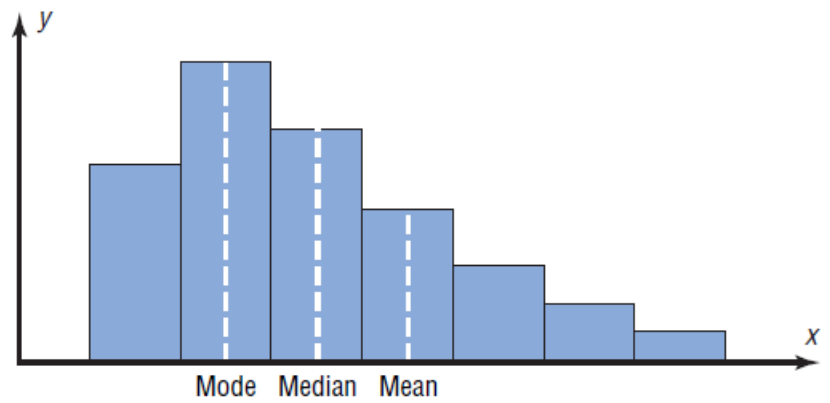
$$\bar{X} = \frac{\sum_{i=1}^n w_iX_i}{\sum_{i=1}^n w_i} = \frac{(3)(4) + (3)(2) + (4)(3) + (2)(1)}{3 + 3 + 4 + 2} = \frac{32}{12} = 2.7$$

The grade point average is 2.7.

Distribution Shapes

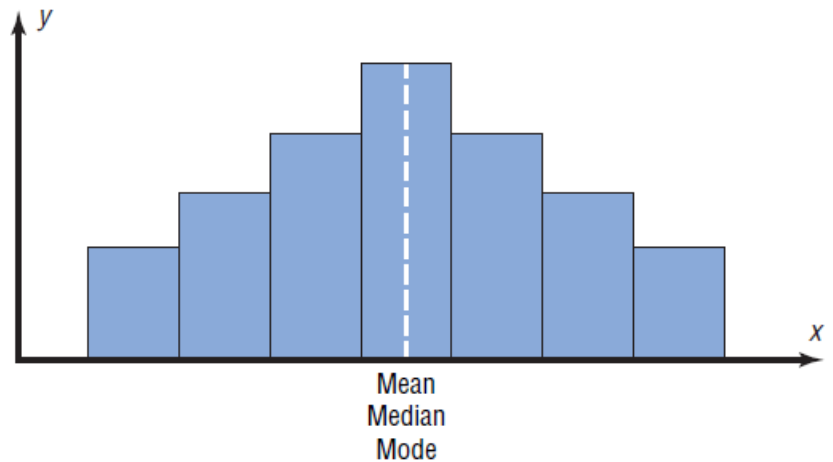
Positively skewed, symmetric, and negatively skewed:

In a **positively skewed** or **right-skewed distribution**, the majority of the data values fall to the left of the mean and cluster at the lower end of the distribution; the “tail” is to the right. Also, the mean is to the right of the median, and the mode is to the left of the median.



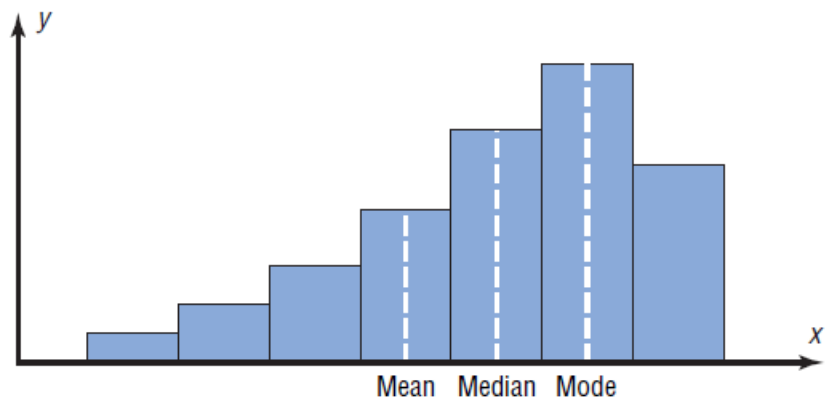
(a) Positively skewed or right-skewed

In a **symmetric distribution**, the data values are evenly distributed on both sides of the mean. In addition, when the distribution is unimodal, the mean, median, and mode are the same and are at the center of the distribution.



(b) Symmetric

When the majority of the data values fall to the right of the mean and cluster at the upper end of the distribution, with the tail to the left, the distribution is said to be **negatively skewed** or **left-skewed**. Also, the mean is to the left of the median, and the mode is to the right of the median.



(c) Negatively skewed or left-skewed